# Mercy House

Evidence-grounded investigations, from an accessible assault to an entangled town inquest.

## World and tasks

Eight tasks, four causal case families, three designed levels. The detective receives its own remembered life and the local scene; it chooses where to go, whom to question and what to inspect. Character agents have isolated lives and memories. Required answers and private plans are never supplied.

| Family / level | Investigation |
|---|---|
| Lantern Room / introductory | Witnessed lamp assault; seeded assailant; four locals, three places. |
| Second Passenger / intermediate | Coach sabotage; two deaths; intended target differs from casualty count. |
| Bell Before Dawn / advanced | Apparently prophetic warning; forged paper, delivery chain and death. |
| Mercy House / intermediate–advanced | Five scopes over one canonical incident; sixteen profiles, fourteen places. |

Canonical reset executes a recorded LLM action tape. New families execute authored seeded action tapes. Preconditions, movement and local perception create persistent traces. Removing sabotage or adulteration prevents the corresponding deaths. Historical background is authored; continuous offstage lives are unfinished.

## Actions and reward

Observe, identify people, travel, search or delegate, secure, ask, seek access, compare, excavate and report. Presence, permissions, custody and types are validated. Characters cannot create evidence through dialogue. A casebook preserves acquired sources and unscored notes.

```
R = clamp(C − 0.20 F − 0.10 U, 0, 1)
```

C: supported scoped target coverage; F/U: distinct false/unsupported findings. Full success requires C=1 and no F/U. A correct name alone earns zero. Authored, programmatic proof routes require acquired provenance; no LLM judge. Narrative is retained but unscored.

## Actual calibration

Four cold-start native-tool detective runs with live character models solved Lantern Room with score 1.0 and no false/unsupported findings, including both culprit assignments. This is a small development set, not a reliability estimate. Other families have scripted feasibility proofs; complete live solving is not established.

# Isolation, evaluation and limits

## Protect truth from the solver

Four unprivileged containers separate world, credentials, trusted harness and real shell. The shell has no verifier, private files, orchestration token, keys or external route. Its gateway accepts only validated public tools. Private character planners receive derived question topics; public speakers see only released memories. Live UI events omit private memories and provider reasoning. Docker security assumes a trusted host/daemon.

## Run contract

Exact Compose command; MODEL/API_KEY/BASE_URL, TASK_ID/SEED/MODE; exact UTF-8 APPEND_FILE suffix. Selected-task reference/no-op gates run before detective calls. Forty replies, ten calls per reply, 600 seconds; tool and grading deadlines. Malformed replies consume steps. Logs retain requests, raw outputs, actions, errors, state hashes, reward, latency and usage. Optional localhost story viewer remains independent.

## Measured checks

All 184 task/seed gate pairs passed (eight tasks, 23 seeds); references 1 and no-ops 0. Six public-only scripted investigations also solved the three new families with both culprit assignments. 120 town, 9 legacy and 9 Node tests pass. Browser/filter/form checks and real-command isolation probes pass. Fresh no-cache four-image build: 26.338 seconds locally; organizer timing remains external.

## Policy training and transfer

Earlier five-skill prototype: 20,000 CPU episodes; unchanged weights rechecked 147/160 with corrected sources. Historical training remains separately labeled. New fresh-zero selector: 6,000 episodes, 132 coefficients, 48.251 seconds, zero training API calls. Three original mechanism-inspired checkpoints, 48 training and 48 held-out states; greedy 96/96 versus untrained 3/96. Prepared acquisitions and rule-based candidate parsing are disclosed; no LLM fine-tuning or full-TV-episode training. New two-scope elite calibration: LLM-ranked 0/2; source-trained selector 0/2. Same API proposal contract and grounded fixed character response bank; not the native live-character harness.

## Known weaknesses

Authored small families and proof vocabulary; small transfer sample; no direct score for internal grief/ego; no discovered supernatural mechanism. Live refusal can block the sole independent named witness in the canonical incident. Other levels are design complexity, not measured success-rate bands. Source originality is authored, not an exhaustive uniqueness guarantee. Final separate NPC setup and upload remain external. No winning or universal jailbreak-resistance claim.

Primary research and PS links: docs/RESEARCH_AND_WIN_PLAN.md. Full trajectories and source manifests accompany this entry.